

NTU SPACE DATA CENTER

SDTwin POWER MODULE

Power Module Software Test Report

Solar Array Battery Response Power Allocation Protection Simulink Comparison

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Version Control

Abstract	This report records the framework, inputs, criteria and individual comparisons of the Power model with independent Simulink simulation. All 20 numerical comparisons pass; 17 cases pass design acceptance and 3 fail due to the full-charge constraint.			
Keywords	Power model; Simulink; test cases; numerical comparison; acceptance records			
Change Record				
Revision	Change Request	Date	Author	Description
1.0	Test execution	2026-10-07		Executed 20 model comparisons with Simulink and retained all results.
1.1	Standard format	2026-10-07		Applied the Orbit seven-chapter layout, case records and landscape summary; data and criteria unchanged.

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1 Overview

1.1 Identification

Platform: SDTwin. Module: Power. Document revision: 1.1. Test date: 7 October 2026.

Purpose: record repeatable comparison tests of the project model against Simulink and the actual acceptance verdicts.

1.2 Software and Document Overview

This report compares the project Power model with Simulink under identical parameters, states and inputs. Tests cover array power, cell voltage and heat, pack scaling, allocation, protection commands, dynamic trajectories, conservation and numerical precision.

Table 1. System Under Test and Reference Implementation

Object	Implementation and boundary
Project model	Thermal/sdtwin_sim/power_stand_in.py
Implementation status	A test implementation of the Power design report, not production EPS software. The source under test is unchanged.
Simulink	Independent MATLAB equations in a Level 2 S-function in a real .slx model, with continuous states integrated by Simulink ode45.
Independence	Only input parameters are shared. MATLAB does not call Python; Python does not take Simulink outputs as its state trajectory.
Parameter provenance	Existing illustrative array and battery assets, including illustrative electrode polynomials, not fitted to a selected physical cell.
Supported conclusion	Cross-implementation verification under common physics assumptions and selected design behavior checks. This is neither measured validation nor a comparison against a factory Simscape battery block.

Observed conclusion

All 20 projects executed and passed numerical comparison. CT-003, DY-002 and DY-007 fail full-charge behavior requirements. Both implementations constrain current, voltage and material fractions but omit the configured 100% usable-charge endpoint from the charging constraint.

Failed cases retain their inputs and raw results. The model under test was not changed, tolerances were not relaxed, and SOC was not clipped to hide the failures.

1.3 Abbreviations and Conventions

Table 2. Abbreviations and Symbols

Term	Definition
SOC	Usable charge fraction: zero is empty and one is full.
PDU	Power distribution unit; losses follow the specified efficiency.
RMSE	Root mean square error at identical samples.
$G / \cos \theta$	Irradiance including eclipse and panel incidence cosine.
i / PB	Positive for discharge, negative for charge.

2 Test Content

Twenty projects form four capability groups. Each maps to one detailed case record and one final summary record. Numerical agreement and design behavior are evaluated and recorded separately.

2.1 Functional Requirement and Test Case Traceability

Table 3. Functional Requirement and Test Case Traceability

No.	Case ID	Function	Test Name	Basis	Status
1	PV-001	solar_power	Solar irradiance and panel attitude	P2	Pass
2	BT-001	battery_response	Open circuit voltage and initial charge	P4–P8, P10	Pass
3	BT-002	battery_response	Charge and discharge voltage versus temperature	P4–P8, P10	Pass
4	BT-003	battery_response	Battery heat and reversible heat sign	P4–P8, P10	Pass
5	BT-004	battery_response	Cell to series and parallel pack scaling	P4–P8, P10	Pass
6	PD-001	solve_power_allocation	Solar and battery power allocation	P1, P3, P11	Pass
7	PD-002	solve_power_allocation	Distribution efficiency and loss	P1, P3, P11	Pass
8	CT-001	solve_power_allocation	Charge limits and solar curtailment	P1, P3, P11	Pass
9	CT-002	solve_power_allocation	Discharge capability and supply shortfall	P1, P3, P11	Pass
10	CT-003	solve_power_allocation	Charging boundary at full state of charge	P1, P3, P11	Fail
11	PR-001	apply_power_event	Protection commands and latched states	5	Pass
12	IV-001	PowerInputs / PowerState	Invalid inputs and state boundaries	5	Pass
13	DY-001	solve_power_allocation / state integration	Charging under continuous sunlight	P1–P11	Pass
14	DY-002	solve_power_allocation / state integration	Eclipse entry and exit	P1–P11	Fail
15	DY-003	solve_power_allocation / state integration	Compute load steps	P1–P11	Pass
16	DY-004	solve_power_allocation / state integration	Dynamic charge curtailment	P1–P11	Pass
17	DY-005	solve_power_allocation / state integration	Supply loss latch and explicit restart	P1–P11	Pass
18	DY-006	solve_power_allocation / state integration	Battery electrothermal feedback	P1–P11	Pass
19	DY-007	solve_power_allocation / state integration	Energy balance over three orbit cycles	P1–P11	Fail
20	NU-001	DOP853 / ode45	Solver convergence and accuracy review	P1–P11	Pass

2.2 Comparison Framework

Table 4. Test Framework and Execution Contract

Stage	Execution contract
Shared inputs	suite.json stores device parameters, initial conditions, input combinations, event times, numerical settings and thresholds, bound by SHA256.
Independent execution	Python calls the existing API with SciPy DOP853 integration; MATLAB independently evaluates cell and allocation equations while Simulink ode45 integrates states.
Events and time	Scheduled discontinuities split integration intervals exactly. Both left and right values are retained at each boundary. Protection is applied only at accepted boundaries.
Signal comparison	Subtract values at the same time, side and unit. No interpolation crosses an event. Invalid results compare validity and event flags without replacing missing ports by zero.
Independent acceptance	In addition to numerical agreement, check power and lithium conservation, solar feasibility, usable SOC range, full-charge stopping and expected protection behavior.
Evidence and report	Retain both CSV trajectories, direct difference CSVs, event JSON, check verdicts and source hashes. Every case includes an input table, error table and comparison figure.

Physical relations and states

$$P_{pv} + P_B = P_{load} + Q_D \quad P_B = N_s N_p i v \quad P_{load} = \eta_D P_{bus}$$

$$dx_n/dt = -i/Q_n \quad dx_p/dt = i/Q_p \quad SOC = (x_n - x_{n,0})/(x_{n,100} - x_{n,0})$$

Positive cell current and battery power denote discharge; negative values denote charging. Q_B is internal battery heat and is not subtracted again from terminal-power balance. Battery temperature is integrated once in the thermal domain.

Electrothermal cases use $CB \, dTB/dt = Q_B - (TB - T_{sink})/R$. Other dynamic cases prescribe a constant temperature and do not run a second temperature model.

3 Detailed Test Projects

Every project uses the Orbit test report record fields. Input tables specify the scenario; result pages retain both outputs, direct differences and error tables. Human signature fields are intentionally blank.

3.1 Solar Array and Battery Components

3.1.1 Test Project 1 Solar irradiance and panel attitude

Table 5. PV-001 Test Case

Case ID	PV-001	Case Name	Solar irradiance and panel attitude
Test Objective	Verify how irradiance and panel orientation determine available power. Zero irradiance denotes full eclipse; a negative incidence cosine places the Sun behind the active face. Both must produce zero power.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P2.		
Input Data	All irradiances are crossed with seven incidence cosines. Python rotates the panel normal with a quaternion; Simulink receives the cosine of the same geometry. An analytical area times efficiency times front-face irradiance check is also applied. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Rear illumination and eclipse yield zero power; other outputs equal active area times efficiency times front-face irradiance.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 21 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 42 checks recorded; evidence: results/PV-001.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	No anomaly against this case criteria.		

PV-001 Scenario Inputs and Initial Conditions

Table 6. PV-001 Scenario Inputs

Index	G W/m ²	cos θ
1	0	-1
2	0	-0.5
3	0	0
4	0	0.25
5	0	0.5
6	0	0.866025
7	0	1
8	680.5	-1
9	680.5	-0.5
10	680.5	0
11	680.5	0.25
12	680.5	0.5
13	680.5	0.866025
14	680.5	1
15	1361	-1
16	1361	-0.5
17	1361	0
18	1361	0.25
19	1361	0.5
20	1361	0.866025
21	1361	1

Parameters not overridden here use sections 5.2 and 6.1.

PV-001 Comparison Results

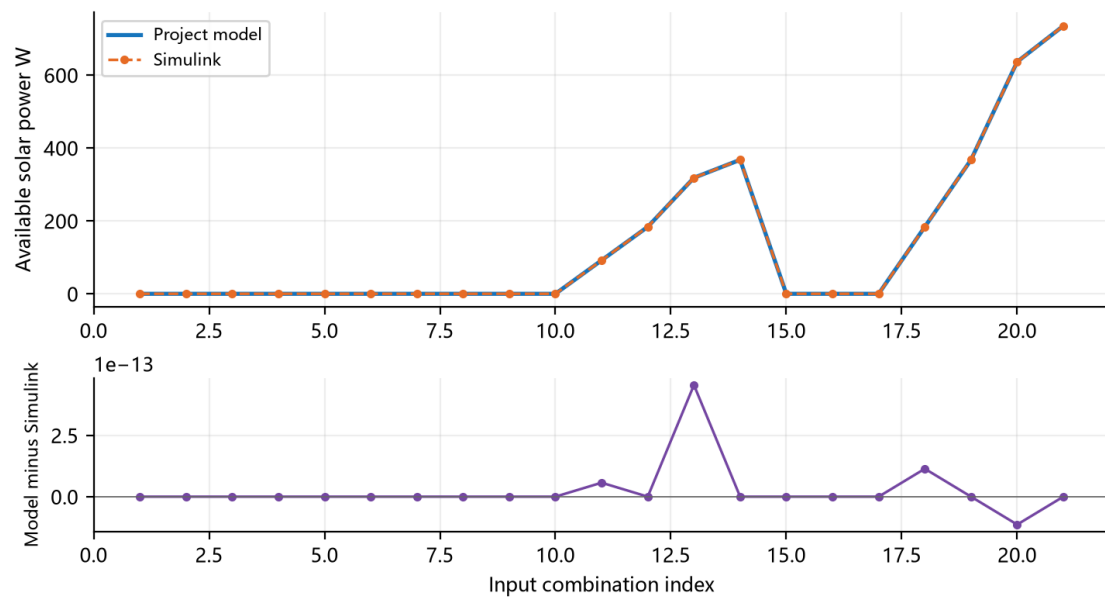


Figure 1 PV-001: Available solar power W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 7. PV-001 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Available solar power W	4.55e-13	1.06e-13	1.00e-06
State of charge fraction	4.44e-16	4.44e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 42 checks.

Completed 21 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.1.2 Test Project 2 Open circuit voltage and initial charge

Table 8. BT-001 Test Case

Case ID	BT-001	Case Name	Open circuit voltage and initial charge
Test Objective	Verify open circuit voltage at different initial charge levels. At zero cell current there is no charge transfer or current-induced heat; terminal voltage is the difference between electrode equilibrium potentials.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P4–P8, P10.		
Input Data	Sweep the negative-electrode mean fraction and derive the positive fraction from lithium inventory conservation. Temperature is 293.15 K or 20 degrees Celsius. The samples are independent states, not a charging trajectory. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Zero current yields zero heat and fraction derivatives; terminal voltage equals the electrode equilibrium-potential difference.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 9 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 58 checks recorded; evidence: results/BT-001.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

BT-001 Scenario Inputs and Initial Conditions

Table 9. BT-001 Scenario Inputs

Index	xn	xp	TB K	i A	Ns × Np
1	0.1	0.773163	293.15	0	8 × 6
2	0.2	0.710267	293.15	0	8 × 6
3	0.3	0.647372	293.15	0	8 × 6
4	0.4	0.584477	293.15	0	8 × 6
5	0.5	0.521581	293.15	0	8 × 6
6	0.6	0.458686	293.15	0	8 × 6
7	0.7	0.395791	293.15	0	8 × 6
8	0.8	0.332895	293.15	0	8 × 6
9	0.88	0.282579	293.15	0	8 × 6

Parameters not overridden here use sections 5.2 and 6.1.

BT-001 Comparison Results

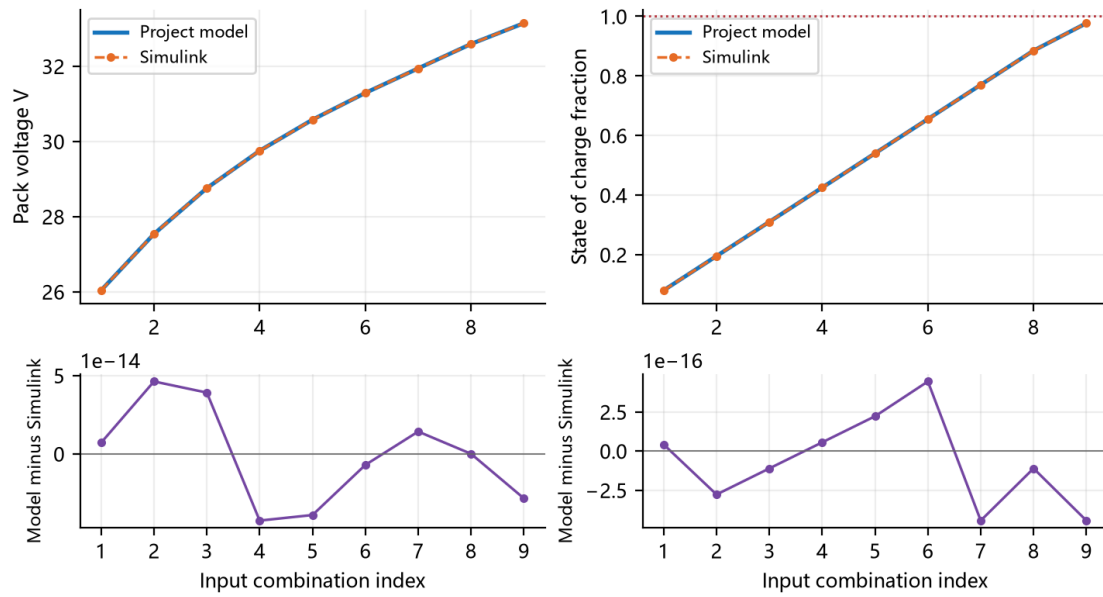


Figure 2 BT-001: Pack voltage V and State of charge fraction. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 10. BT-001 Numerical Errors and Limits

Signal	Max error	RMSE	Limit
Pack voltage V	4.62e-14	3.00e-14	1.00e-04
State of charge fraction	4.44e-16	2.88e-16	1.00e-06
Pack current A	0	0	1.00e-04
Pack heat W	0	0	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 58 checks.

Completed 9 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.1.3 Test Project 3 Charge and discharge voltage versus temperature

Table 11. BT-002 Test Case

Case ID	BT-002	Case Name	Charge and discharge voltage versus temperature
Test Objective	Verify charge and discharge voltage response and the effect of temperature on kinetics and diffusion. Positive current is discharge and negative current is charge. Temperature is the battery temperature, not ambient air in space.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P4–P8, P10.		
Input Data	Cross three cell temperatures with six cell currents at a common lithium state. Direct cell evaluation retains negative constraint margins for boundary diagnostics; not every sample is a feasible power-allocation operating point. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Voltage, polarization and margins agree; infeasible current samples retain diagnostics and are not labeled feasible supply states.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 18 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 58 checks recorded; evidence: results/BT-002.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

BT-002 Scenario Inputs and Initial Conditions

Table 12. BT-002 Scenario Inputs

Index	xn	xp	TB K	i A	Ns × Np
1	0.6	0.458686	273.15	−1	8 × 6
2	0.6	0.458686	273.15	−0.5	8 × 6
3	0.6	0.458686	273.15	0	8 × 6
4	0.6	0.458686	273.15	0.5	8 × 6
5	0.6	0.458686	273.15	1	8 × 6
6	0.6	0.458686	273.15	2	8 × 6
7	0.6	0.458686	293.15	−1	8 × 6
8	0.6	0.458686	293.15	−0.5	8 × 6
9	0.6	0.458686	293.15	0	8 × 6
10	0.6	0.458686	293.15	0.5	8 × 6
11	0.6	0.458686	293.15	1	8 × 6
12	0.6	0.458686	293.15	2	8 × 6
13	0.6	0.458686	313.15	−1	8 × 6
14	0.6	0.458686	313.15	−0.5	8 × 6
15	0.6	0.458686	313.15	0	8 × 6
16	0.6	0.458686	313.15	0.5	8 × 6
17	0.6	0.458686	313.15	1	8 × 6
18	0.6	0.458686	313.15	2	8 × 6

Parameters not overridden here use sections 5.2 and 6.1.

BT-002 Comparison Results

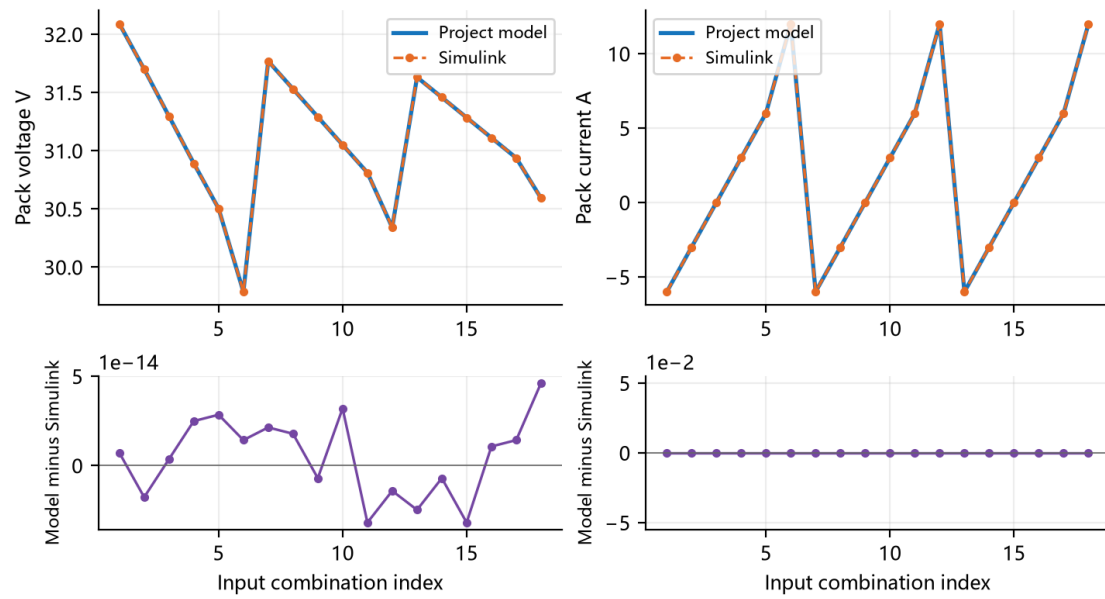


Figure 3 BT-002: Pack voltage V and Pack current A. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 13. BT-002 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Pack voltage V	4.62e-14	2.26e-14	1.00e-04
Pack current A	0	0	1.00e-04
Pack heat W	3.20e-14	7.78e-15	1.00e-06
State of charge fraction	4.44e-16	4.44e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 58 checks.

Completed 18 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.1.4 Test Project 4 Battery heat and reversible heat sign

Table 14. BT-003 Test Case

Case ID	BT-003	Case Name	Battery heat and reversible heat sign
Test Objective	Verify reaction heat, ohmic heat and signed reversible heat. Reversible heat can make total heat negative at low current; this denotes net heat absorption in the model and must not be clipped to zero.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P4–P8, P10.		
Input Data	Hold both lithium fractions fixed and scan small positive and negative currents at 283.15 K and 303.15 K. Compare pack heat and cell reversible heat separately. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Reaction, ohmic and reversible heat sum consistently; negative reversible heat and total heat are retained.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Pack heat ranges from -0.0320434 to 3.23576 W; negative heat is retained. 60 checks recorded; evidence: results/BT-003.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

BT-003 Scenario Inputs and Initial Conditions

Table 15. BT-003 Scenario Inputs

Index	xn	xp	TB K	i A	Ns × Np
1	0.6	0.458686	283.15	−0.5	8 × 6
2	0.6	0.458686	283.15	−0.1	8 × 6
3	0.6	0.458686	283.15	0	8 × 6
4	0.6	0.458686	283.15	0.1	8 × 6
5	0.6	0.458686	283.15	0.5	8 × 6
6	0.6	0.458686	283.15	1	8 × 6
7	0.6	0.458686	303.15	−0.5	8 × 6
8	0.6	0.458686	303.15	−0.1	8 × 6
9	0.6	0.458686	303.15	0	8 × 6
10	0.6	0.458686	303.15	0.1	8 × 6
11	0.6	0.458686	303.15	0.5	8 × 6
12	0.6	0.458686	303.15	1	8 × 6

Parameters not overridden here use sections 5.2 and 6.1.

BT-003 Comparison Results

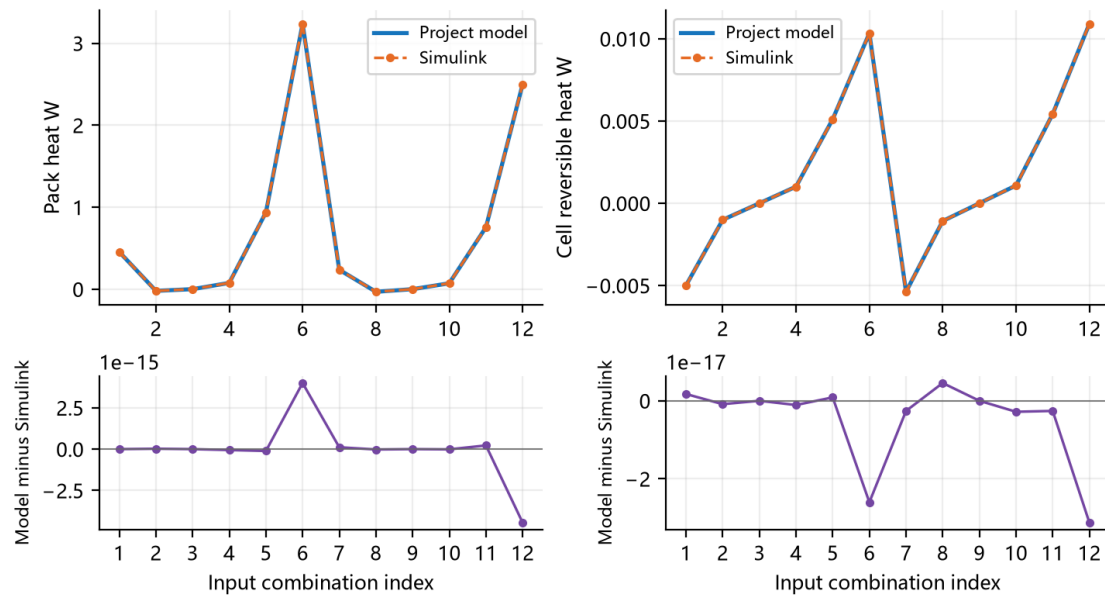


Figure 4 BT-003: Pack heat W and Cell reversible heat W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 16. BT-003 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Pack heat W	4.44e-15	1.73e-15	1.00e-06
Cell reversible heat W	3.12e-17	1.19e-17	1.00e-06
Pack voltage V	4.26e-14	2.32e-14	1.00e-04
Pack current A	1.11e-16	6.41e-17	1.00e-04
State of charge fraction	4.44e-16	4.44e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 60 checks.

Pack heat ranges from -0.0320434 to 3.23576 W; negative heat is retained.

3.1.5 Test Project 5 Cell to series and parallel pack scaling

Table 17. BT-004 Test Case

Case ID	BT-004	Case Name	Cell to series and parallel pack scaling
Test Objective	Verify cell-to-pack conversion. Series count scales pack voltage, parallel count scales pack current, and their product scales pack power and heat. Scaling must be applied once.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P4–P8, P10.		
Input Data	Five series and parallel configurations share the same cell state, 0.5 A cell discharge current and 293.15 K temperature. Cells have identical states and temperature, with equal parallel current sharing. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Pack voltage scales with series count, current with parallel count, and power and heat with total cell count.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 5 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 58 checks recorded; evidence: results/BT-004.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

BT-004 Scenario Inputs and Initial Conditions

Table 18. BT-004 Scenario Inputs

Index	xn	xp	TB K	i A	Ns × Np
1	0.6	0.458686	293.15	0.5	1 × 1
2	0.6	0.458686	293.15	0.5	8 × 1
3	0.6	0.458686	293.15	0.5	1 × 6
4	0.6	0.458686	293.15	0.5	8 × 6
5	0.6	0.458686	293.15	0.5	12 × 4

Parameters not overridden here use sections 5.2 and 6.1.

BT-004 Comparison Results

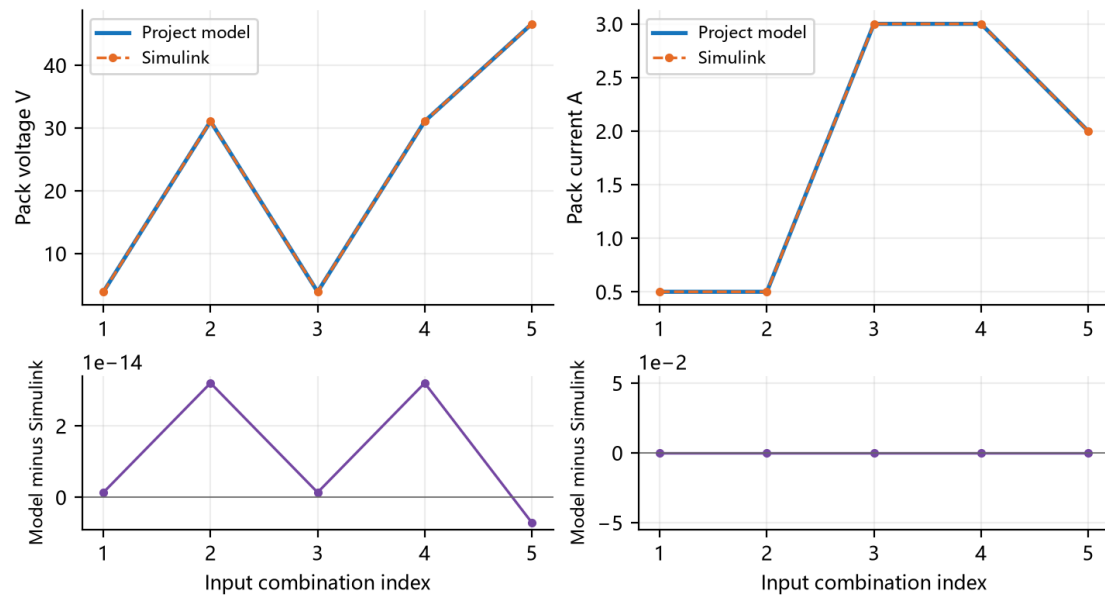


Figure 5 BT-004: Pack voltage V and Pack current A. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 19. BT-004 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Pack voltage V	3.20e-14	2.05e-14	1.00e-04
Pack current A	0	0	1.00e-04
Pack heat W	3.33e-16	2.46e-16	1.00e-06
State of charge fraction	4.44e-16	4.44e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 58 checks.

Completed 5 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.2 Power Allocation and Protection Interfaces

3.2.1 Test Project 6 Solar and battery power allocation

Table 20. PD-001 Test Case

Case ID	PD-001	Case Name	Solar and battery power allocation
Test Objective	Verify power allocation among array, battery and load. Surplus solar charges the battery; a deficit calls for battery discharge. A request beyond the allowed battery capability must report supply shortfall.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1, P3, P11.		
Input Data	Cross four irradiances with six load requests. Distribution efficiency is 0.95, so bus-side demand exceeds load demand. Shortfall samples compare status flags and must not publish artificial zero heat as a valid result. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Surplus solar charges the battery; a deficit discharges it; infeasible requests explicitly report shortfall.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 24 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 57 checks recorded; evidence: results/PD-001.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

PD-001 Scenario Inputs and Initial Conditions

Table 21. PD-001 Scenario Inputs

Index	G W/m ²	Demand W	TB K	Override
1	0	0	293.15	Baseline
2	0	50	293.15	Baseline
3	0	200	293.15	Baseline
4	0	400	293.15	Baseline
5	0	900	293.15	Baseline
6	0	2500	293.15	Baseline
7	300	0	293.15	Baseline
8	300	50	293.15	Baseline
9	300	200	293.15	Baseline
10	300	400	293.15	Baseline
11	300	900	293.15	Baseline
12	300	2500	293.15	Baseline
13	800	0	293.15	Baseline
14	800	50	293.15	Baseline
15	800	200	293.15	Baseline
16	800	400	293.15	Baseline
17	800	900	293.15	Baseline
18	800	2500	293.15	Baseline
19	1361	0	293.15	Baseline
20	1361	50	293.15	Baseline
21	1361	200	293.15	Baseline
22	1361	400	293.15	Baseline
23	1361	900	293.15	Baseline
24	1361	2500	293.15	Baseline

Parameters not overridden here use sections 5.2 and 6.1.

PD-001 Comparison Results

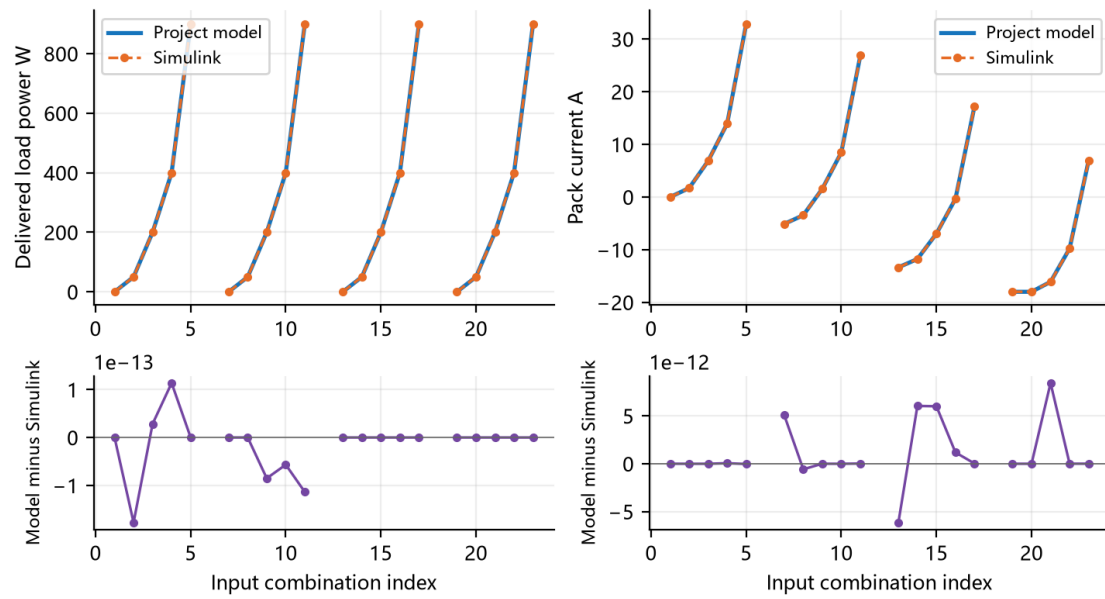


Figure 6 PD-001: Delivered load power W and Pack current A. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 22. PD-001 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Delivered load power W	1.78e-13	5.86e-14	1.00e-06
Pack current A	8.43e-12	3.23e-12	1.00e-04
Actual solar power W	2.85e-10	1.07e-10	1.00e-06
Pack voltage V	5.97e-13	2.46e-13	1.00e-04
Pack heat W	1.58e-11	4.70e-12	1.00e-06
State of charge fraction	4.44e-16	4.44e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 57 checks.

Completed 24 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.2.2 Test Project 7 Distribution efficiency and loss

Table 23. PD-002 Test Case

Case ID	PD-002	Case Name	Distribution efficiency and loss
Test Objective	Verify that distribution input splits into load power and distribution heat, with no omitted or double-counted loss.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1, P3, P11.		
Input Data	Set solar input to zero and supply the load from the battery at efficiencies 0.80, 0.95 and 1.00. The ideal 1.00 boundary must have zero distribution heat. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Load plus distribution heat equals distribution input; unit efficiency produces zero distribution heat.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 9 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 57 checks recorded; evidence: results/PD-002.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

PD-002 Scenario Inputs and Initial Conditions

Table 24. PD-002 Scenario Inputs

Index	G W/m ²	Demand W	TB K	Override
1	0	0	293.15	eta=0.8
2	0	100	293.15	eta=0.8
3	0	300	293.15	eta=0.8
4	0	0	293.15	eta=0.95
5	0	100	293.15	eta=0.95
6	0	300	293.15	eta=0.95
7	0	0	293.15	eta=1
8	0	100	293.15	eta=1
9	0	300	293.15	eta=1

Parameters not overridden here use sections 5.2 and 6.1.

PD-002 Comparison Results

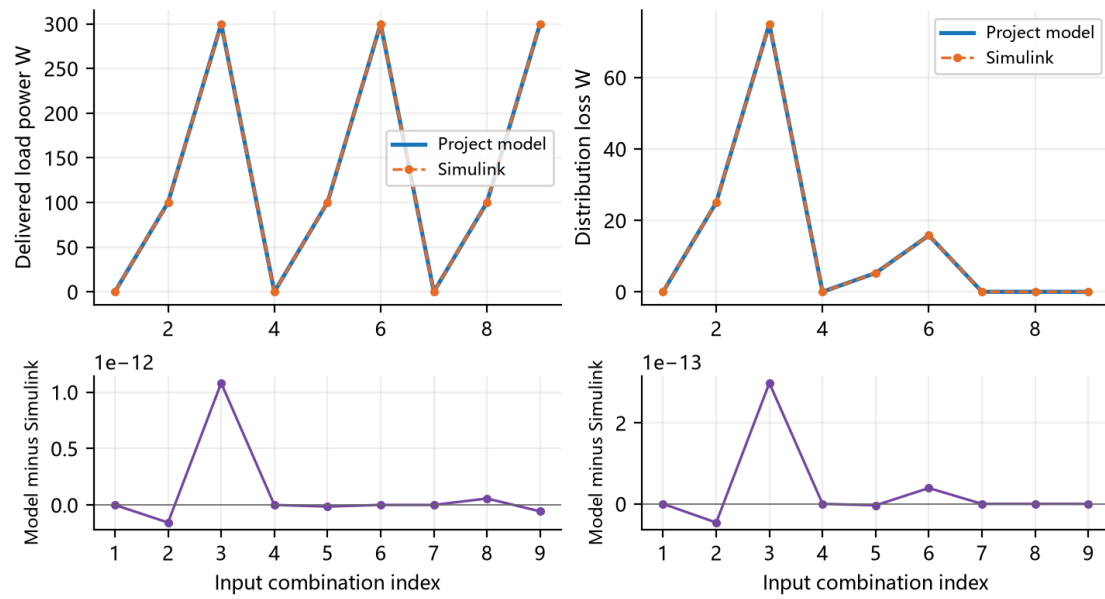


Figure 7 PD-002: Delivered load power W and Distribution loss W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 25. PD-002 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Delivered load power W	1.08e-12	3.65e-13	1.00e-06
Distribution loss W	2.98e-13	1.02e-13	1.00e-06
Actual solar power W	0	0	1.00e-06
Pack voltage V	4.26e-14	2.15e-14	1.00e-04
Pack current A	4.44e-14	1.68e-14	1.00e-04
Pack heat W	4.62e-14	1.55e-14	1.00e-06
State of charge fraction	4.44e-16	4.44e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 57 checks.

Completed 9 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.2.3 Test Project 8 Charge limits and solar curtailment

Table 26. CT-001 Test Case

Case ID	CT-001	Case Name	Charge limits and solar curtailment
Test Objective	Verify curtailment when solar surplus exceeds the battery charge capability. Current, voltage and particle center or surface fractions can constrain charging.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1, P3, P11.		
Input Data	Use an artificial 1800 W/m ² irradiance and a 25 W load to guarantee surplus. This is a boundary stress test, not a solar constant assigned to a real orbit. Tighten charge current or voltage limits, or lower battery temperature. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Honor charge limits and curtail solar while satisfying feasible load and terminal-power conservation.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 6 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 57 checks recorded; evidence: results/CT-001.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

CT-001 Scenario Inputs and Initial Conditions

Table 27. CT-001 Scenario Inputs

Index	G W/m ²	Demand W	TB K	Override
1	1800	25	293.15	imin=-0.2
2	1800	25	293.15	vmax=4.03
3	1800	25	253.15	Baseline
4	1800	25	273.15	Baseline
5	1800	25	313.15	Baseline
6	1800	25	293.15	Baseline

Parameters not overridden here use sections 5.2 and 6.1.

CT-001 Comparison Results

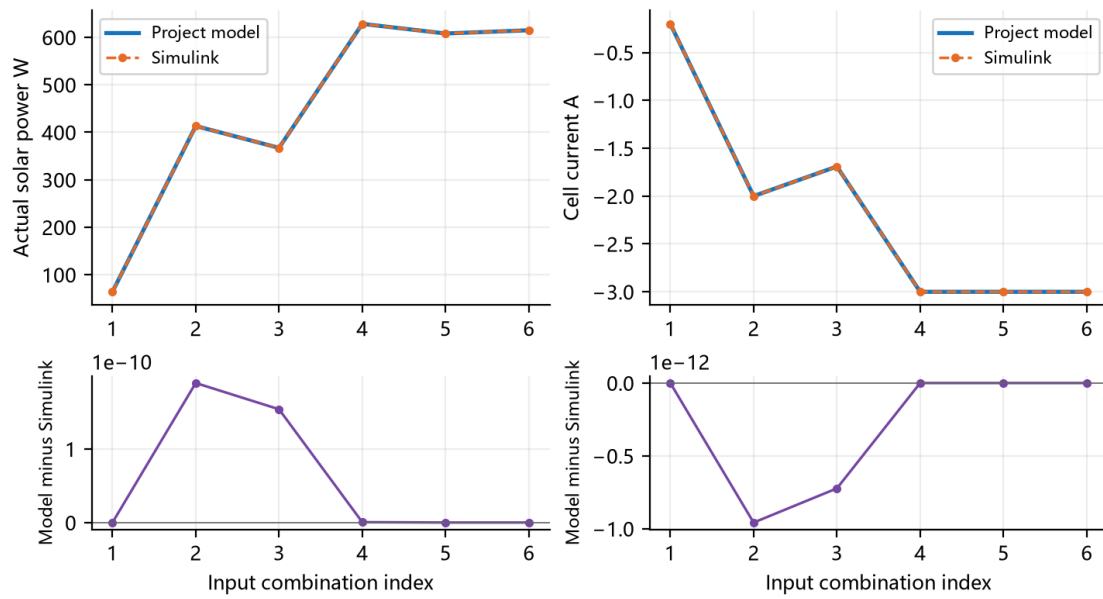


Figure 8 CT-001: Actual solar power W and Cell current A. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 28. CT-001 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Actual solar power W	1.90e-10	9.97e-11	1.00e-06
Cell current A	9.58e-13	4.90e-13	1.00e-04
Delivered load power W	0	0	1.00e-06
Pack voltage V	7.60e-13	3.59e-13	1.00e-04
Pack current A	5.75e-12	2.94e-12	1.00e-04
Pack heat W	1.02e-11	5.30e-12	1.00e-06
State of charge fraction	4.44e-16	4.44e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 57 checks.

Completed 6 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.2.4 Test Project 9 Discharge capability and supply shortfall

Table 29. CT-002 Test Case

Case ID	CT-002	Case Name	Discharge capability and supply shortfall
Test Objective	Verify discharge limits and supply shortfall classification. A physically infeasible request must request an event and must not be mislabeled as a numerical solver failure.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1, P3, P11.		
Input Data	With no solar input, cross three requests with four battery settings: baseline, reduced discharge current, higher minimum voltage and low temperature. Each sample solves its own allowed current interval. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Infeasible demand returns shortfall and an event request, distinguished from numerical failure.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 12 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 57 checks recorded; evidence: results/CT-002.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

CT-002 Scenario Inputs and Initial Conditions

Table 30. CT-002 Scenario Inputs

Index	G W/m ²	Demand W	TB K	Override
1	0	20	293.15	Baseline
2	0	100	293.15	Baseline
3	0	1000	293.15	Baseline
4	0	20	293.15	imax=0.25
5	0	100	293.15	imax=0.25
6	0	1000	293.15	imax=0.25
7	0	20	293.15	vmin=3.9
8	0	100	293.15	vmin=3.9
9	0	1000	293.15	vmin=3.9
10	0	20	253.15	Baseline
11	0	100	253.15	Baseline
12	0	1000	253.15	Baseline

Parameters not overridden here use sections 5.2 and 6.1.

CT-002 Comparison Results

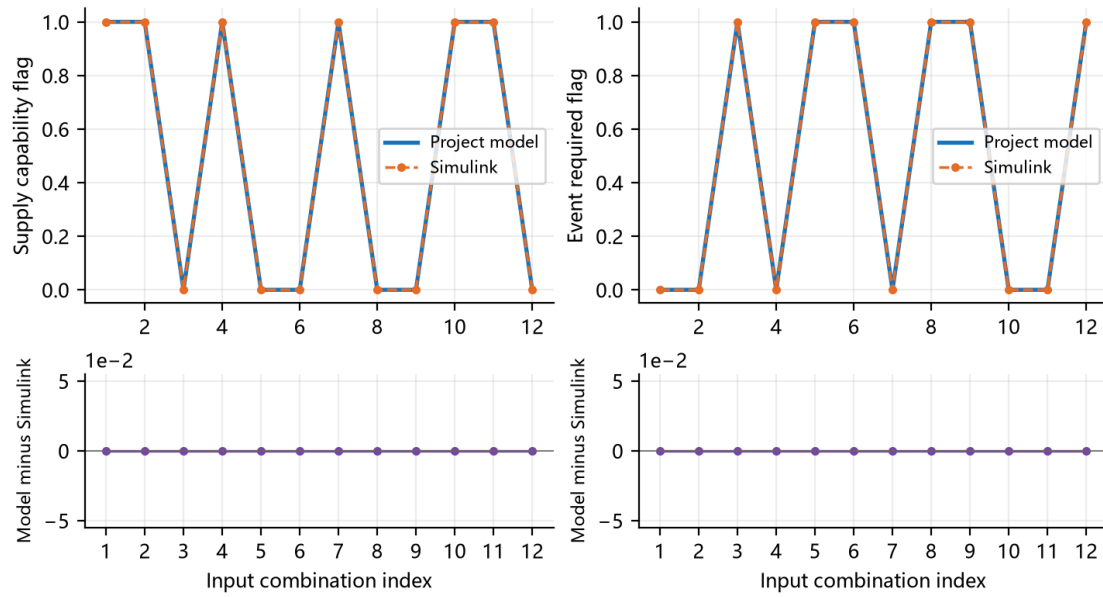


Figure 9 CT-002: Supply capability flag and Event required flag. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 31. CT-002 Numerical Errors and Limits

Signal	Max error	absolute	RMSE	Limit
Supply capability flag	0		0	0
Event required flag	0		0	0
Actual solar power W	0		0	1.00e-06
Delivered load power W	1.11e-10		4.52e-11	1.00e-06
Pack voltage V	2.59e-13		1.11e-13	1.00e-04
Pack current A	3.74e-12		1.53e-12	1.00e-04
Pack heat W	6.34e-13		2.59e-13	1.00e-06
State of charge fraction	4.44e-16		4.44e-16	1.00e-06

Numerical comparison passes; overall acceptance: Pass. Recorded 57 checks.

Completed 12 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.2.5 Test Project 10 Charging boundary at full state of charge

Table 32. CT-003 Test Case

Case ID	CT-003	Case Name	Charging boundary at full state of charge
Test Objective	Verify the design requirement to limit charging at full charge. The configured 100% SOC point is negative fraction 0.90 and positive fraction 0.27. Solar surplus at this state should not continue charging the battery.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1, P3, P11.		
Input Data	Set initial SOC to 1, request 25 W, and apply three irradiances. Coverage review added this case after orbital runs exceeded SOC 1; the original numerical comparison tolerances were not relaxed. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	At SOC one, stop further charging and curtail surplus solar.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	All three full-charge samples continue charging, with minimum cell current -0.544984 A. Numerical agreement passes, but the full-charge requirement fails. 59 checks recorded; evidence: results/CT-003.json.		
Recorded By		Test Conclusion	☐ Pass ☒ Fail ☐ Not Executed
Anomalies	All three full-charge samples continue charging, with minimum cell current -0.544984 A. Numerical agreement passes, but the full-charge requirement fails.		

CT-003 Scenario Inputs and Initial Conditions

Table 33. CT-003 Scenario Inputs

Index	G W/m ²	Demand W	TB K	Override
1	800	25	293.15	Baseline
2	1361	25	293.15	Baseline
3	1800	25	293.15	Baseline

$x_n = 0.90$; $x_p = 0.27$; SOC = 1; $\cos \theta = 1$.

Parameters not overridden here use sections 5.2 and 6.1.

CT-003 Comparison Results

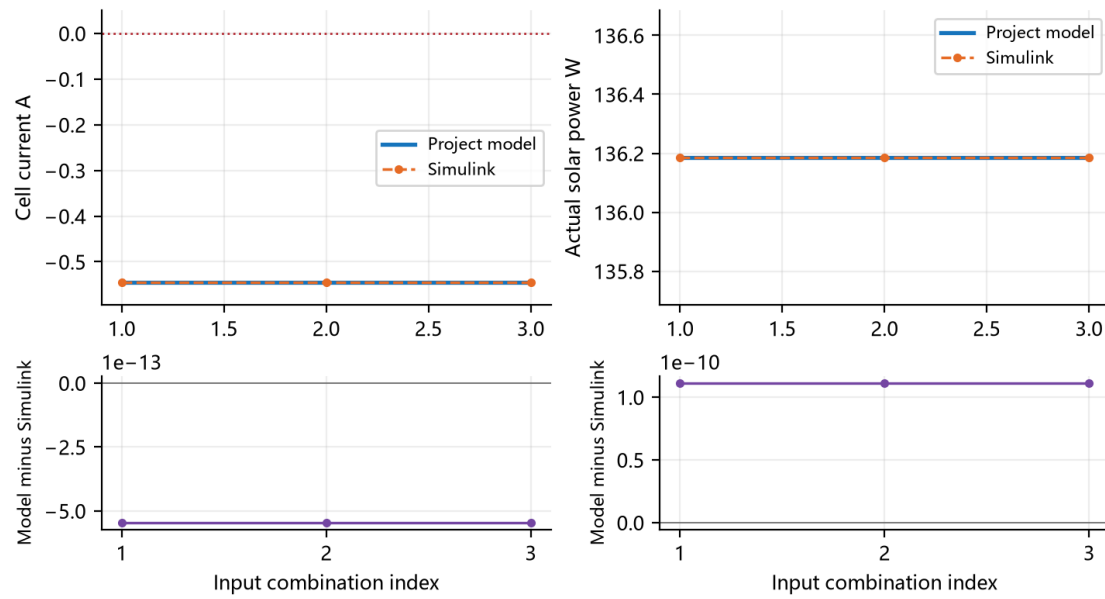


Figure 10 CT-003: Cell current A and Actual solar power W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 34. CT-003 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Cell current A	5.46e-13	5.46e-13	1.00e-04
Actual solar power W	1.11e-10	1.11e-10	1.00e-06
Delivered load power W	0	0	1.00e-06
Pack voltage V	2.98e-13	2.98e-13	1.00e-04
Pack current A	3.27e-12	3.27e-12	1.00e-04
Pack heat W	1.48e-12	1.48e-12	1.00e-06
State of charge fraction	0	0	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Fail. Recorded 59 checks.

All three full-charge samples continue charging, with minimum cell current -0.544984 A.

Numerical agreement passes, but the full-charge requirement fails.

3.2.6 Test Project 11 Protection commands and latched states

Table 35. PR-001 Test Case

Case ID	PR-001	Case Name	Protection commands and latched states
Test Objective	Verify start and stop commands, loss-of-supply latching, denied restart, simultaneous command priority and duplicate commands. Restored sunlight alone must not restart a latched load.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: 5.		
Input Data	Eight configurations cover stop, feasible and infeasible start, supply loss, simultaneous stop and start, duplicate stop, and an unaccepted trial event. A trial event must be rejected without changing state. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Start, stop, latching, command priority and trial-event rejection match the prescribed event sequence.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 8 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 59 checks recorded; evidence: results/PR-001.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

PR-001 Scenario Inputs and Initial Conditions

Table 36. PR-001 Scenario Inputs

Index	G W/m ²	Demand W	Connected / latched	Command
1	1361	200	1 / 0	stop
2	1361	200	0 / 0	start
3	0	2500	0 / 1	start
4	1361	200	0 / 1	start
5	0	2500	1 / 0	supply_loss
6	1361	200	1 / 0	stop_start
7	1361	200	1 / 0	repeat_stop
8	0	2500	1 / 0	trial_supply_loss

Parameters not overridden here use sections 5.2 and 6.1.

PR-001 Comparison Results

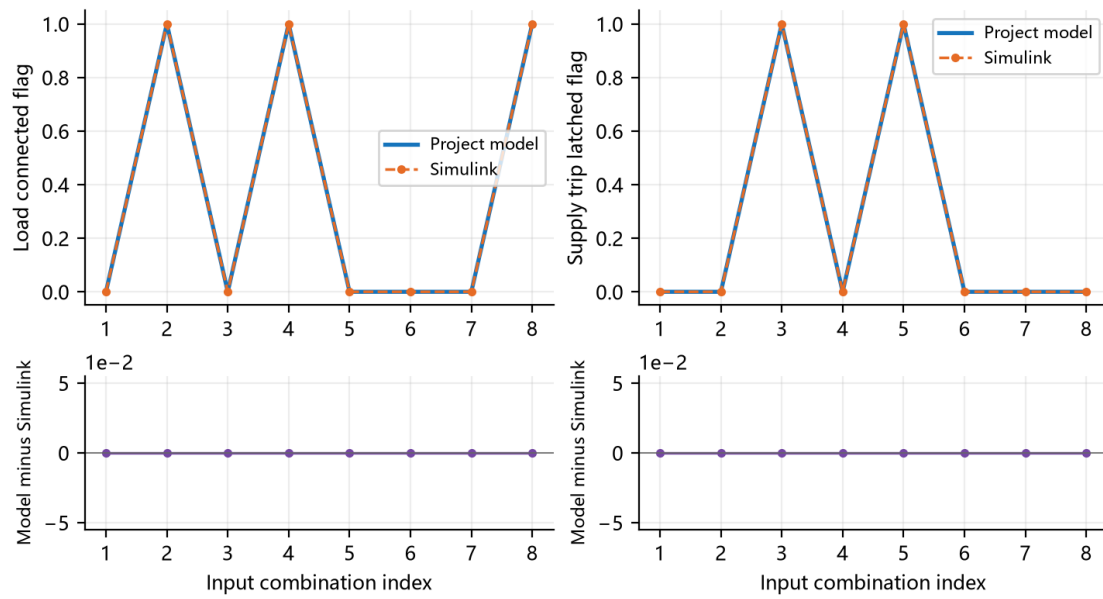


Figure 11 PR-001: Load connected flag and Supply trip latched flag. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 37. PR-001 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Load connected flag	0	0	0
Supply trip latched flag	0	0	0
Actual solar power W	2.85e-10	1.52e-10	1.00e-06
Delivered load power W	0	0	1.00e-06
Pack voltage V	5.97e-13	3.19e-13	1.00e-04
Pack current A	8.43e-12	4.51e-12	1.00e-04
Pack heat W	1.58e-11	8.45e-12	1.00e-06
State of charge fraction	4.44e-16	4.44e-16	1.00e-06

Numerical comparison passes; overall acceptance: Pass. Recorded 59 checks.

Completed 8 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.2.7 Test Project 12 Invalid inputs and state boundaries

Table 38. IV-001 Test Case

Case ID	IV-001	Case Name	Invalid inputs and state boundaries
Test Objective	Distinguish invalid input from a physical state boundary. Invalid input must not drive protection; a trial state outside material limits must report a device boundary requiring handling.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: 5.		
Input Data	Test 240 K battery temperature, mismatched run identity, mismatched time, unsupported frame, zero Sun direction and an out-of-range mean lithium fraction. The first five are invalid; the last is a valid boundary-event report. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	The first five inputs are invalid; the sixth is a valid device-boundary event. Invalid results do not fabricate port values.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Completed 6 input combinations. Output availability, discrete status and applicable numerical difference checks pass. 40 checks recorded; evidence: results/IV-001.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

IV-001 Scenario Inputs and Initial Conditions

Table 39. IV-001 Scenario Inputs

Index	Input change	Expected class
1	temperature	Invalid input
2	identity	Invalid input
3	time	Invalid input
4	frame	Invalid input
5	zero_sun	Invalid input
6	mean_boundary	Boundary event

Parameters not overridden here use sections 5.2 and 6.1.

IV-001 Comparison Results

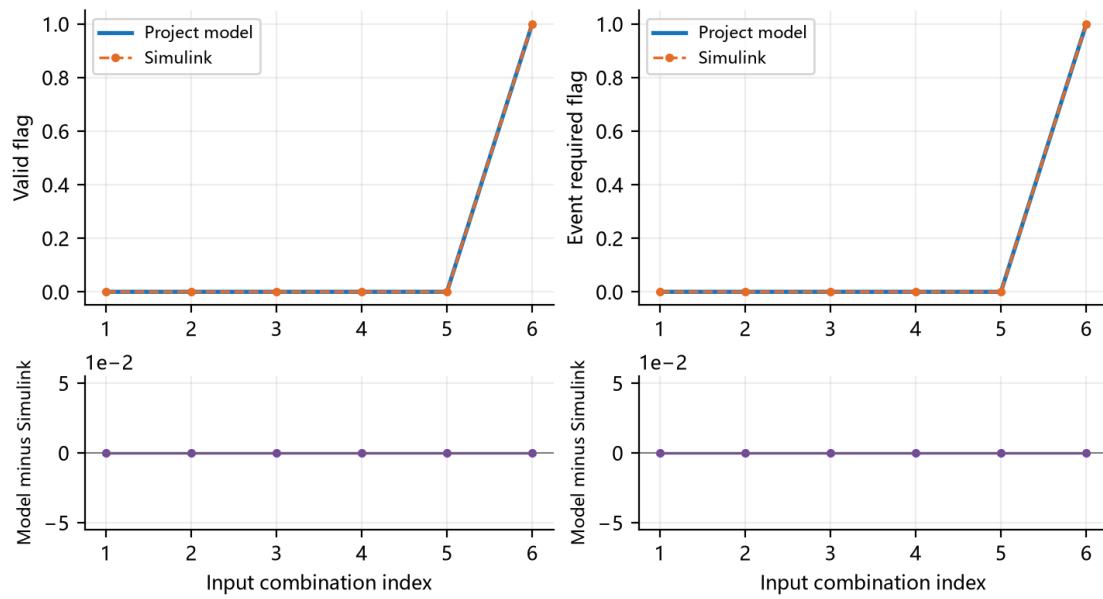


Figure 12 IV-001: Valid flag and Event required flag. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

Horizontal indices match the input table for this case. Connecting lines aid reading and do not imply a dynamic trajectory between independent samples.

Table 40. IV-001 Numerical Errors and Limits

Signal	Max error	absolute	RMSE	Limit
Valid flag	0		0	0
Event required flag	0		0	0
State of charge fraction	4.44e-16		4.44e-16	1.00e-06

Numerical comparison passes; overall acceptance: Pass. Recorded 40 checks.

Completed 6 input combinations. Output availability, discrete status and applicable numerical difference checks pass.

3.3 Dynamic Supply and Electrothermal Processes

3.3.1 Test Project 13 Charging under continuous sunlight

Table 41. DY-001 Test Case

Case ID	DY-001	Case Name	Charging under continuous sunlight
Test Objective	Verify gradual charging in continuous sunlight and agreement of SOC, current, voltage and heat between the independent integrators.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	Normal solar incidence at 1361 W/m ² , a 200 W load and a prescribed 293.15 K battery temperature for 900 s. Save results every 10 s. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	SOC rises continuously within zero to one, power is conserved, and dynamic outputs agree within tolerance.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	SOC changes from 65.517% to 88.654%; battery temperature changes from 293.150 K to 293.150 K. 87 checks recorded; evidence: results/DY-001.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

DY-001 Scenario Inputs and Initial Conditions

Table 42. DY-001 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 900	1361	200	none

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 293.15 K; incidence cosine 1; output interval 10 s.

Parameters not overridden here use sections 5.2 and 6.1.

DY-001 Comparison Results

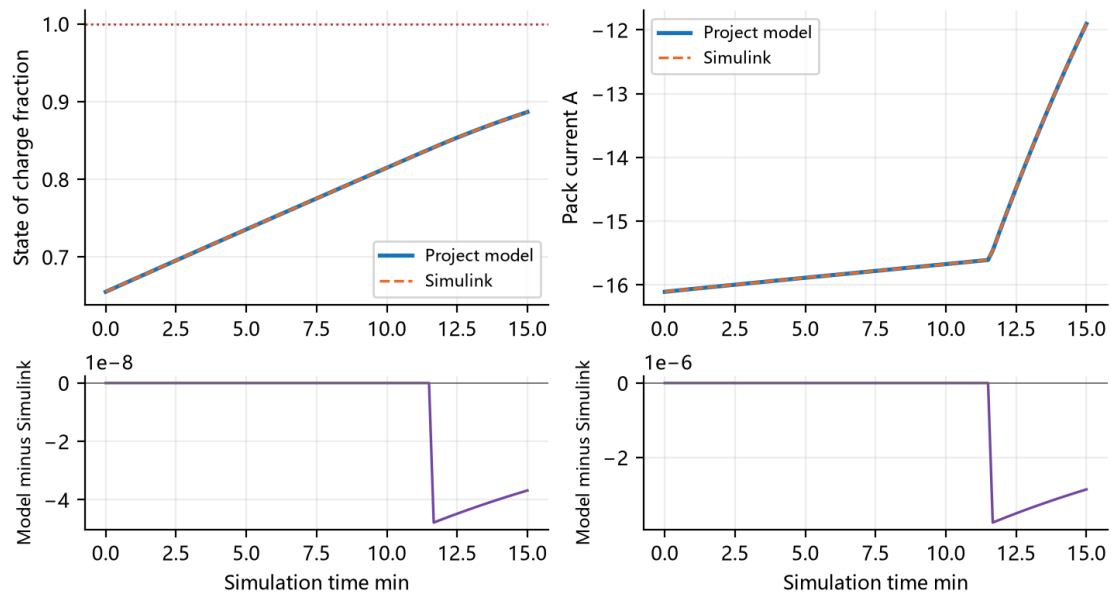


Figure 13 DY-001: State of charge fraction and Pack current A. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 43. DY-001 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
State of charge fraction	4.78e-08	2.03e-08	1.00e-06
Pack current A	3.75e-06	1.58e-06	1.00e-04
Actual solar power W	1.26e-04	5.32e-05	0.001
Delivered load power W	0	0	0.001
Pack voltage V	1.57e-12	7.88e-13	1.00e-04
Pack heat W	6.81e-06	2.61e-06	0.001
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 87 checks.

SOC changes from 65.517% to 88.654%; battery temperature changes from 293.150 K to 293.150 K.

3.3.2 Test Project 14 Eclipse entry and exit

Table 44. DY-002 Test Case

Case ID	DY-002	Case Name	Eclipse entry and exit
Test Objective	Verify transitions from sunlight into eclipse and back. Solar output is zero in eclipse and the battery supplies the load; sunlight recovery restores array power and possible charging.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	Use a controlled 5400 s cycle with full eclipse from 1800 to 3800 s and 1361 W/m ² otherwise. Load stays at 200 W. This fixture is neither ISS telemetry nor orbit-propagator output. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Battery supplies the eclipse interval; array supply resumes in sunlight; SOC stays within zero to one.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Peak SOC is 100.6153%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails. 87 checks recorded; evidence: results/DY-002.json.		
Recorded By		Test Conclusion	☐ Pass ☒ Fail ☐ Not Executed
Anomalies	Peak SOC is 100.6153%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails.		

DY-002 Scenario Inputs and Initial Conditions

Table 45. DY-002 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 1800	1361	200	none
1800 to 3800	0	200	none
3800 to 5400	1361	200	none

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 293.15 K; incidence cosine 1; output interval 20 s.

Parameters not overridden here use sections 5.2 and 6.1.

DY-002 Comparison Results

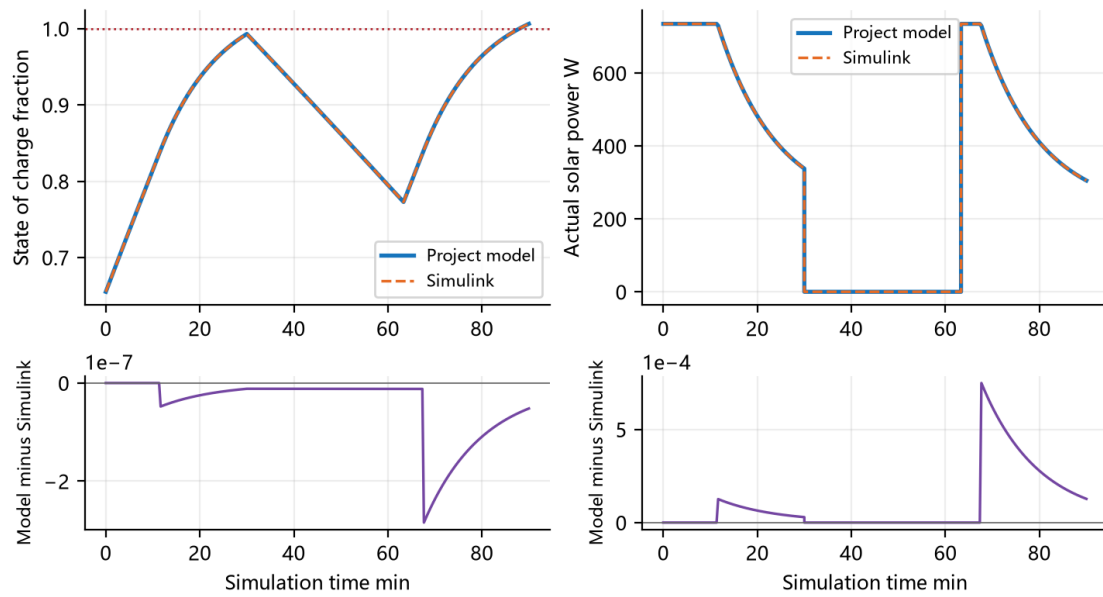


Figure 14 DY-002: State of charge fraction and Actual solar power W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 46. DY-002 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
State of charge fraction	2.84e-07	7.69e-08	1.00e-06
Actual solar power W	7.49e-04	1.98e-04	0.001
Delivered load power W	1.89e-10	2.35e-11	0.001
Pack voltage V	7.15e-08	4.38e-08	1.00e-04
Pack current A	2.23e-05	5.90e-06	1.00e-04
Pack heat W	4.04e-05	8.05e-06	0.001
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Fail. Recorded 87 checks.

Peak SOC is 100.6153%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails.

3.3.3 Test Project 15 Compute load steps

Table 47. DY-003 Test Case

Case ID	DY-003	Case Name	Compute load steps
Test Objective	Verify battery compensation and charge or discharge transitions when compute demand changes. A step is a sudden change of requested power, not a switching-waveform experiment.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	Hold irradiance at 800 W/m ² and request 100, 350, 550 and 100 W in consecutive 200 s intervals. Temperature is fixed. Segment boundaries retain both pre-step and post-step values. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Power and current respond to demand steps while lithium states remain continuous at each event.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	SOC changes from 65.517% to 71.390%; battery temperature changes from 293.150 K to 293.150 K. 87 checks recorded; evidence: results/DY-003.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

DY-003 Scenario Inputs and Initial Conditions

Table 48. DY-003 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 200	800	100	none
200 to 400	800	350	none
400 to 600	800	550	none
600 to 800	800	100	none

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 293.15 K; incidence cosine 1; output interval 5 s.

Parameters not overridden here use sections 5.2 and 6.1.

DY-003 Comparison Results

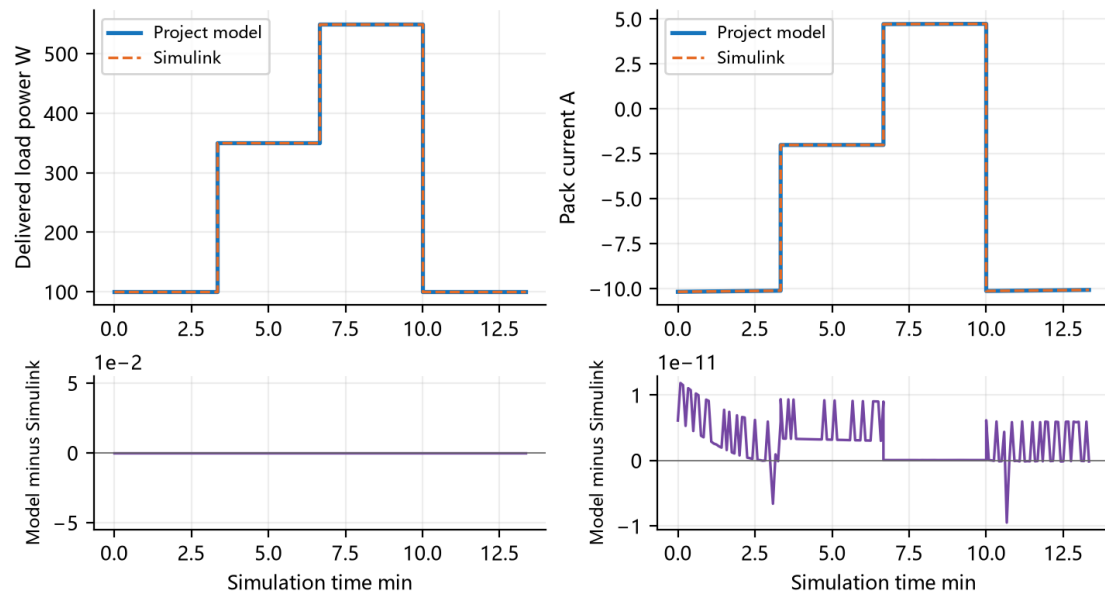


Figure 15 DY-003: Delivered load power W and Pack current A. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 49. DY-003 Numerical Errors and Limits

Signal	Max error	absolute	RMSE	Limit
Delivered load power W	0		0	0.001
Pack current A	1.18e-11		4.50e-12	1.00e-04
Actual solar power W	3.88e-10		1.47e-10	0.001
Pack voltage V	1.13e-12		5.61e-13	1.00e-04
Pack heat W	1.40e-11		4.34e-12	0.001
State of charge fraction	6.98e-14		5.17e-14	1.00e-06
Battery temperature K	0		0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 87 checks.

SOC changes from 65.517% to 71.390%; battery temperature changes from 293.150 K to 293.150 K.

3.3.4 Test Project 16 Dynamic charge curtailment

Table 50. DY-004 Test Case

Case ID	DY-004	Case Name	Dynamic charge curtailment
Test Objective	Verify actual solar output and battery heat as the battery state evolves under a charge current limit.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	Run for 900 s at 1800 W/m ² and 25 W demand, with cell current bounded below by minus 0.2 A. Other parameters are unchanged; the limit applies to a cell, not the pack current. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Charging respects limits, curtailment follows charge capability, and both trajectories agree.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	SOC changes from 65.517% to 67.326%; battery temperature changes from 293.150 K to 293.150 K. 87 checks recorded; evidence: results/DY-004.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	No anomaly against this case criteria.		

DY-004 Scenario Inputs and Initial Conditions

Table 51. DY-004 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 900	1800	25	none

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 293.15 K; incidence cosine 1; output interval 10 s.

Overrides $imin = -0.2$

Parameters not overridden here use sections 5.2 and 6.1.

DY-004 Comparison Results

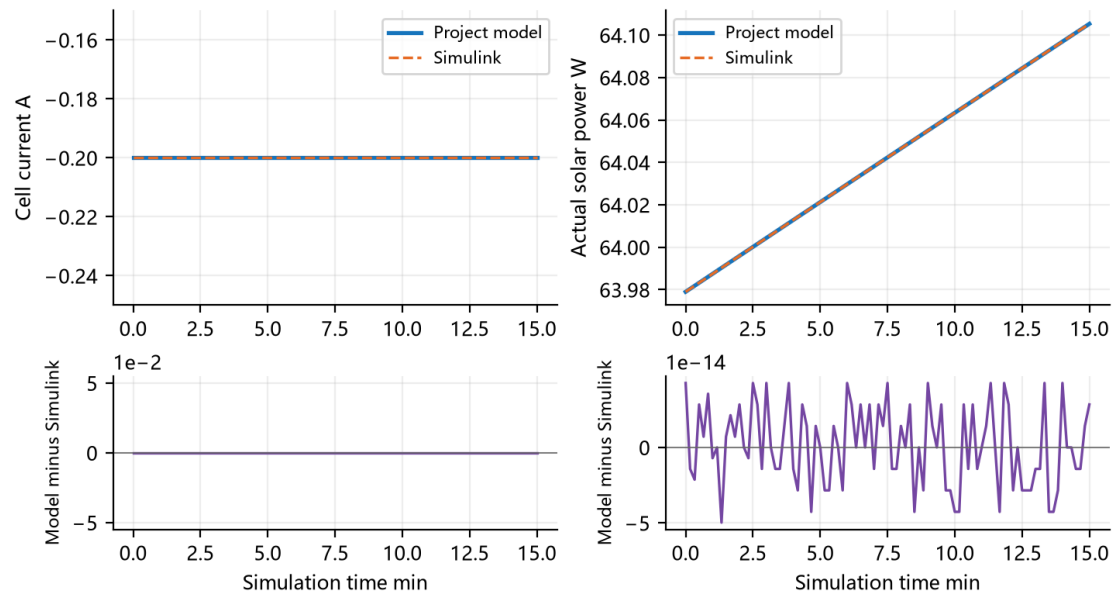


Figure 16 DY-004: Cell current A and Actual solar power W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 52. DY-004 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Cell current A	0	0	1.00e-04
Actual solar power W	4.97e-14	2.64e-14	0.001
Delivered load power W	0	0	0.001
Pack voltage V	4.97e-14	2.77e-14	1.00e-04
Pack current A	0	0	1.00e-04
Pack heat W	7.89e-17	1.52e-17	0.001
State of charge fraction	4.44e-16	2.88e-16	1.00e-06
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 87 checks.

SOC changes from 65.517% to 67.326%; battery temperature changes from 293.150 K to 293.150 K.

3.3.5 Test Project 17 Supply loss latch and explicit restart

Table 53. DY-005 Test Case

Case ID	DY-005	Case Name	Supply loss latch and explicit restart
Test Objective	Verify supply loss, latching, sunlight recovery, explicit restart and a subsequent stop. Protection must change load connectivity while allowing charging with the load disconnected.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	At 100 s remove sunlight and request 2500 W. Restore normal sunlight at 200 s without a start command; start at 300 s, stop at 400 s and start again at 500 s. End at 600 s. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Trip at 100 s; sunlight at 200 s causes no restart; start at 300 s, stop at 400 s, and restart at 500 s.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Supply trips at 100 s; the load remains disconnected after sunlight returns at 200 s. Explicit start at 300 s, stop at 400 s and restart at 500 s agree with expectations. 88 checks recorded; evidence: results/DY-005.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	No anomaly against this case criteria.		

DY-005 Scenario Inputs and Initial Conditions

Table 54. DY-005 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 100	1361	200	none
100 to 200	0	2500	none
200 to 300	1361	200	none
300 to 400	1361	200	start
400 to 500	1361	200	stop
500 to 600	1361	200	start

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 293.15 K; incidence cosine 1; output interval 5 s.

Parameters not overridden here use sections 5.2 and 6.1.

DY-005 Comparison Results

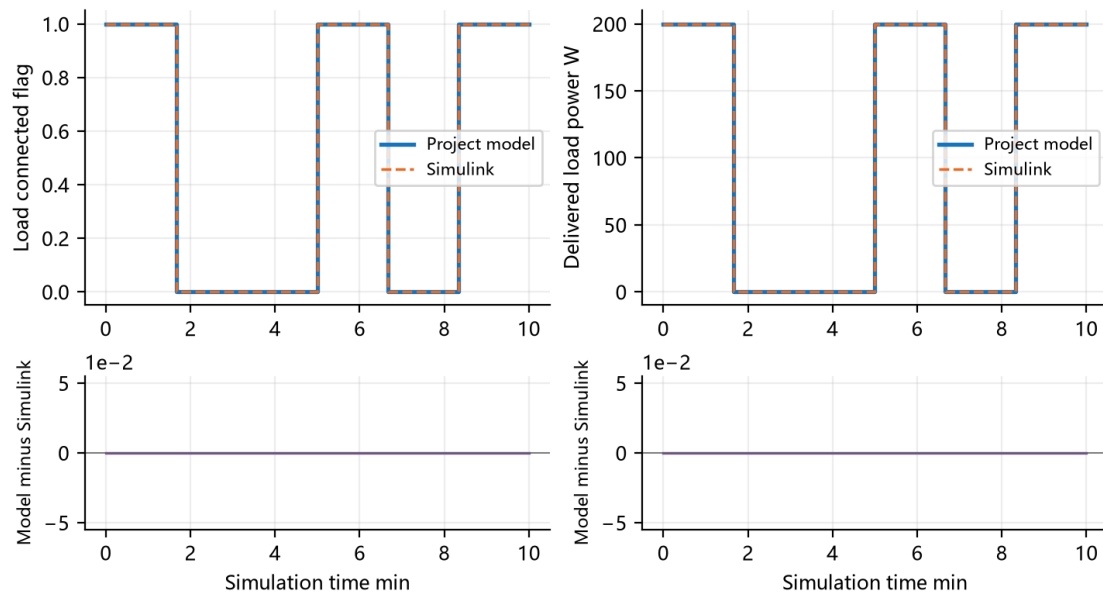


Figure 17 DY-005: Load connected flag and Delivered load power W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 55. DY-005 Numerical Errors and Limits

Signal	Max error	absolute	RMSE	Limit
Load connected flag	0		0	0
Delivered load power W	0		0	0.001
Actual solar power W	6.55e-10		1.57e-10	0.001
Pack voltage V	1.44e-12		3.47e-13	1.00e-04
Pack current A	1.92e-11		4.61e-12	1.00e-04
Pack heat W	3.63e-11		8.72e-12	0.001
State of charge fraction	1.93e-14		6.18e-15	1.00e-06
Battery temperature K	0		0	0.001

Numerical comparison passes; overall acceptance: Pass. Recorded 88 checks.

Supply trips at 100 s; the load remains disconnected after sunlight returns at 200 s. Explicit start at 300 s, stop at 400 s and restart at 500 s agree with expectations.

3.3.6 Test Project 18 Battery electrothermal feedback

Table 56. DY-006 Test Case

Case ID	DY-006	Case Name	Battery electrothermal feedback
Test Objective	Verify the loop in which battery heat changes temperature, which in turn changes voltage and current. The thermal boundary is a prescribed heat-sink temperature, not ambient-air convection.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	Use no solar power, a 200 W load, initial battery temperature 283.15 K, heat capacity 5000 J/K, thermal resistance 0.8 K/W and a 273.15 K sink for 1800 s. This is a one-node battery thermal test, not full six-node Thermal integration. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Temperature is integrated once; heat generation and rejection determine temperature and feed back into voltage and current.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	SOC changes from 65.517% to 44.282%; battery temperature changes from 283.150 K to 280.874 K. 85 checks recorded; evidence: results/DY-006.json.		
Recorded By		Test Conclusion	[X] Pass [] Fail [] Not Executed
Anomalies	No anomaly against this case criteria.		

DY-006 Scenario Inputs and Initial Conditions

Table 57. DY-006 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 1800	0	200	none

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 283.15 K; incidence cosine 1; output interval 10 s.

CB = 5000 J/K; R = 0.8 K/W; $T_{\text{sink}} = 273.15$ K.

Parameters not overridden here use sections 5.2 and 6.1.

DY-006 Comparison Results

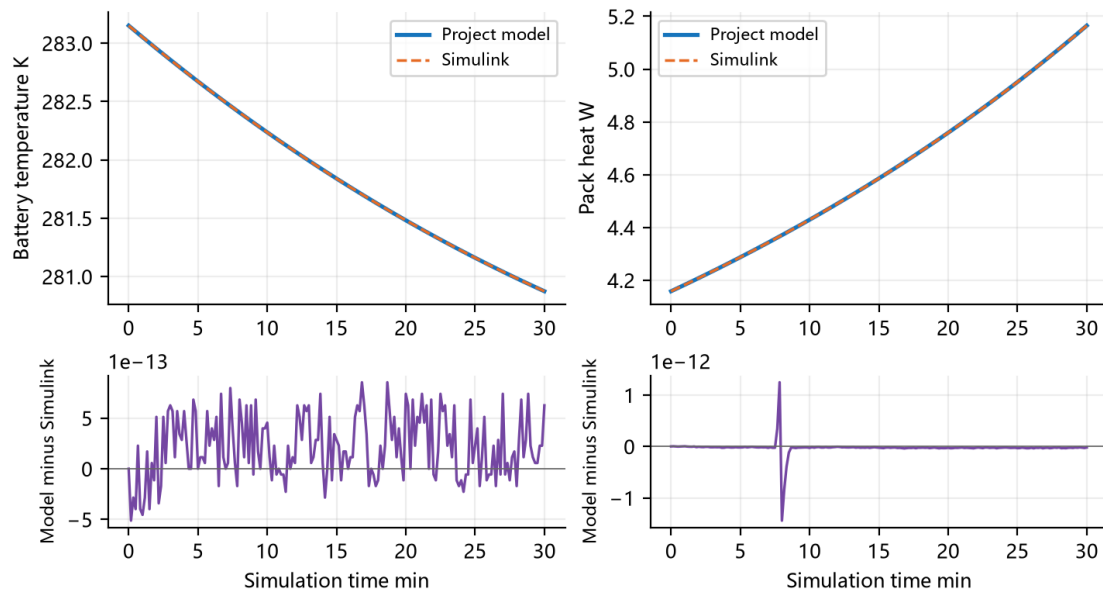


Figure 18 DY-006: Battery temperature K and Pack heat W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 58. DY-006 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Battery temperature K	8.53e-13	3.79e-13	0.001
Pack heat W	1.44e-12	1.62e-13	0.001
Actual solar power W	0	0	0.001
Delivered load power W	3.48e-11	3.90e-12	0.001
Pack voltage V	8.53e-14	3.20e-14	1.00e-04
Pack current A	1.24e-12	1.39e-13	1.00e-04
State of charge fraction	9.99e-16	4.34e-16	1.00e-06

Numerical comparison passes; overall acceptance: Pass. Recorded 85 checks.

SOC changes from 65.517% to 44.282%; battery temperature changes from 283.150 K to 280.874 K.

3.3.7 Test Project 19 Energy balance over three orbit cycles

Table 59. DY-007 Test Case

Case ID	DY-007	Case Name	Energy balance over three orbit cycles
Test Objective	Verify cumulative energy and state drift over repeated sunlight and eclipse cycles. Long trajectories can expose missing charge boundaries despite correct instantaneous power balance.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	Repeat the controlled DY-002 cycle three times for 16200 s. Save every 30 s and retain both sides of all eclipse entry and exit boundaries. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Conservation holds over three cycles, integrated energies agree, and SOC stays within zero to one.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	Peak SOC is 104.2070%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails. 87 checks recorded; evidence: results/DY-007.json.		
Recorded By		Test Conclusion	☐ Pass ☒ Fail ☐ Not Executed
Anomalies	Peak SOC is 104.2070%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails.		

DY-007 Scenario Inputs and Initial Conditions

Table 60. DY-007 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 1800	1361	200	none
1800 to 3800	0	200	none
3800 to 5400	1361	200	none
5400 to 7200	1361	200	none
7200 to 9200	0	200	none
9200 to 10800	1361	200	none
10800 to 12600	1361	200	none
12600 to 14600	0	200	none
14600 to 16200	1361	200	none

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 293.15 K; incidence cosine 1; output interval 30 s.

Parameters not overridden here use sections 5.2 and 6.1.

DY-007 Comparison Results

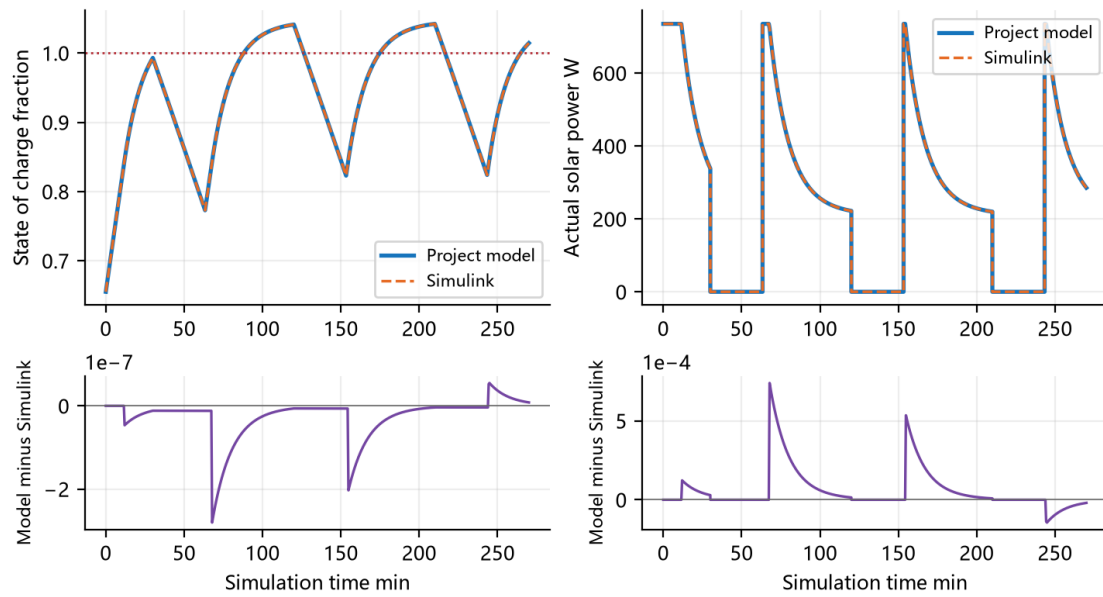


Figure 19 DY-007: State of charge fraction and Actual solar power W. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 61. DY-007 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
State of charge fraction	2.81e-07	5.56e-08	1.00e-06
Actual solar power W	7.39e-04	1.43e-04	0.001
Delivered load power W	1.80e-10	2.08e-11	0.001
Pack voltage V	7.15e-08	2.95e-08	1.00e-04
Pack current A	2.20e-05	4.26e-06	1.00e-04
Pack heat W	3.94e-05	5.66e-06	0.001
Battery temperature K	0	0	0.001

Numerical comparison passes; overall acceptance: Fail. Recorded 87 checks.

Peak SOC is 104.2070%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails.

3.4 Numerical Accuracy Review

3.4.1 Test Project 20 Solver convergence and accuracy review

Table 62. NU-001 Test Case

Case ID	NU-001	Case Name	Solver convergence and accuracy review
Test Objective	Check whether numerical precision explains the observed difference. Independently tighten both solvers and reduce the maximum step, requiring changes below one tenth of the original acceptance limit.		
Preconditions	Use the frozen section 5 inputs and environment; execute both implementations independently. Design basis: P1–P11.		
Input Data	Run 1200 s with no solar input, a 200 W load, initial battery temperature 293.15 K and a 283.15 K sink. Tighten relative tolerance from 10 to the power minus 8 to 10 to the power minus 10 and reduce maximum step from 10 s to 2 s. Exact values appear in the input table for this case.		
Procedure	1. Load this case and run the project model and Simulink independently. 2. Save outputs and events in input order or at the prescribed times. 3. Check validity, same-sample differences and case behavior; retain check records.		
Expected Results	Each solver changes by less than one tenth of the corresponding comparison limit under independent refinement.		
Acceptance Criteria	All applicable signals meet section 5.1 limits, flags and events match expectations, and the expected results in this form hold.		
Case Designer		Executor / Date	Automated test 2026-10-07
Actual Results	SOC changes from 65.517% to 51.553%; battery temperature changes from 293.150 K to 291.345 K. 95 checks recorded; evidence: results/NU-001.json.		
Recorded By		Test Conclusion	☒ Pass ☐ Fail ☐ Not Executed
Anomalies	No anomaly against this case criteria.		

NU-001 Scenario Inputs and Initial Conditions

Table 63. NU-001 Scenario Inputs

Time s	G W/m ²	Demand W	Start command
0 to 1200	0	200	none

Initial $x_n = 0.60$, $x_p = 0.458686001$; initial temperature 293.15 K; incidence cosine 1; output interval 10 s.

CB = 5000 J/K; R = 0.8 K/W; $T_{\text{sink}} = 283.15$ K.

Parameters not overridden here use sections 5.2 and 6.1.

NU-001 Comparison Results

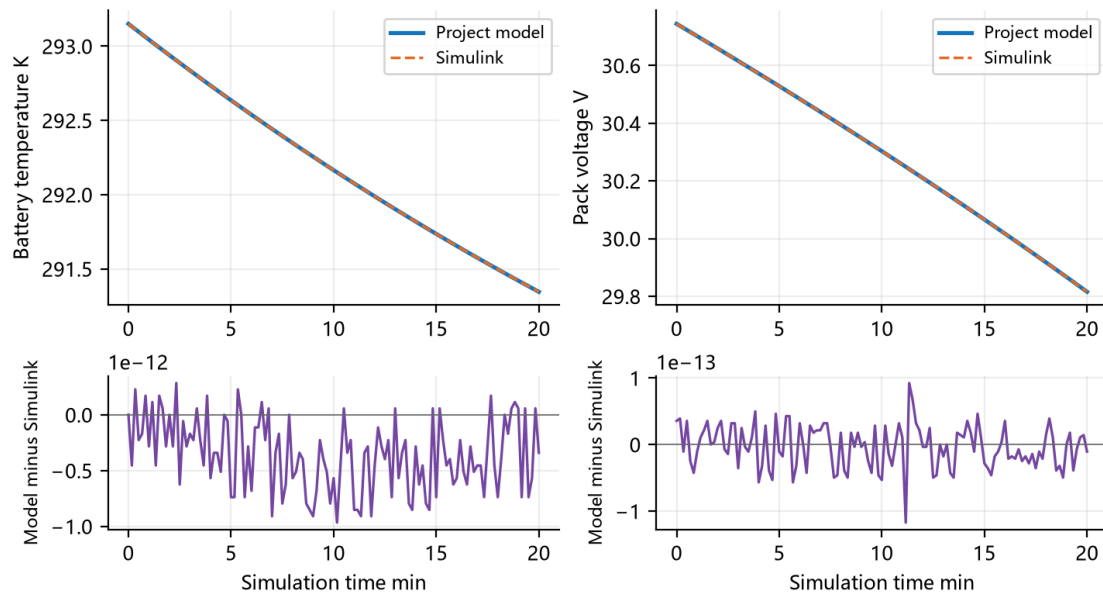


Figure 20 NU-001: Battery temperature K and Pack voltage V. Top panels show both outputs at identical samples; lower panels show project model minus Simulink in the same units.

The horizontal axis is minutes. Vertical changes retain both sides of a discontinuity at the same time; no interpolation spans that discontinuity.

Table 64. NU-001 Numerical Errors and Limits

Signal	Max absolute error	RMSE	Limit
Battery temperature K	9.66e-13	4.79e-13	0.001
Pack voltage V	1.17e-13	3.25e-14	1.00e-04
Actual solar power W	0	0	0.001
Delivered load power W	3.43e-11	4.32e-12	0.001
Pack current A	1.22e-12	1.53e-13	1.00e-04
Pack heat W	1.18e-12	1.51e-13	0.001
State of charge fraction	1.11e-15	4.88e-16	1.00e-06

Numerical comparison passes; overall acceptance: Pass. Recorded 95 checks.

SOC changes from 65.517% to 51.553%; battery temperature changes from 293.150 K to 291.345 K.

4 Test Content Sufficiency Analysis

Table 65. Test Coverage

Capability	Cases	Verified Focus
Solar Array and Battery Components	PV-001, BT-001, BT-002, BT-003, BT-004	Component response, boundary inputs and independent comparison
Power Allocation and Protection Interfaces	PD-001, PD-002, CT-001, CT-002, CT-003, PR-001, IV-001	Component response, boundary inputs and independent comparison
Dynamic Supply and Electrothermal Processes	DY-001, DY-002, DY-003, DY-004, DY-005, DY-006, DY-007	Dynamic states, events, energy or convergence checks
Numerical Accuracy Review	NU-001	Dynamic states, events, energy or convergence checks

This report verifies implementations of the same equations and the listed design behaviors. It does not cover measured-cell accuracy, rapid diffusion relaxation, high-rate electrolyte effects, switching ripple, MPPT tracking transients, autonomous event localization during continuous depletion, full six-node Thermal integration, or every SimReady asset error. Scheduled-boundary protection is covered; this is not evidence of all continuous-boundary localization behavior.

All 20 cases executed, but coverage does not mean all acceptance criteria passed. CT-003, DY-002 and DY-007 require correction of full-charge constraints and retesting. Physical accuracy and broader system integration need additional tests.

5 Test Conditions and Requirements

5.1 Execution Environment and Acceptance Criteria

Table 66. Numerical and Behavioral Acceptance Criteria

Quantity	Maximum absolute difference
Static power and heat	0.000001 W
Dynamic power and heat	0.001 W
Voltage	0.0001 V
Current	0.0001 A
SOC and lithium fractions	0.000001
Battery temperature	0.001 K
Lithium fraction derivative	0.00000001 s ⁻¹
Discrete flags and reason codes	Exact agreement
Scheduled event time	0.001 s
Relative integrated energy difference	0.00001
Power conservation residual	0.000002 W
Relative lithium inventory drift	0.0000000001
Usable SOC range	0 to 1, with a 0.000001 rounding allowance
Full-charge boundary	At SOC 1, cell current must be at least minus 0.00000001 A

Numerical thresholds assess software consistency, not measured cell accuracy. RMSE is reported alongside maximum error; every applicable signal must meet its maximum-error limit. Refinement changes must be below one tenth of the corresponding limit.

Input version 1.0 was frozen before initial execution. Coverage review added CT-003 and SOC behavior criteria in version 1.1 after identifying missing full-charge coverage. Numerical thresholds remain unchanged; prior inputs and criteria are retained in results/history.

MATLAB R2026b 26.2.0.3386108; Simulink 26.2; Windows; Python Thermal/.venv; SciPy DOP853.

Baseline integration uses relative tolerance 10^{-8} , absolute tolerance 10^{-10} and maximum step 10 s. Refinement uses 10^{-10} , 10^{-12} and 2 s.

5.2 Baseline Scenario and Input Requirements

Table 67. Baseline Parameters and Physical Meaning

Input	Value	Meaning
A	1.8 m ²	Active solar cell area
η_{pv}	0.30	Fixed maximum-power-point efficiency
G	1361 W/m ²	Nominal illumination; zero denotes eclipse. Eclipse is already included and is not multiplied twice.
$\cos \theta$	1	Normal incidence; zero is edge-on and negative is rear-face illumination.
$N_s \times N_p$	8×6	Eight series cells per string and six parallel strings, totaling 48 identical cells.
η_D	0.95	Efficiency of distribution and its load-side harness

Input	Value	Meaning
$R\Omega$	0.025 Ω	Cell ohmic resistance
TB	293.15 K / 20 °C	Initial or prescribed battery temperature, not space ambient air.
x_n / x_p	0.600000 / 0.458686001	Mean electrode lithium fractions, dimensionless
SOC	65.517241%	Usable state of charge derived from negative fraction
$x_{n,0} / x_{n,100} / x_{p,100}$	0.03 / 0.90 / 0.27	Empty and full-charge endpoints from the same parameter set
Cell current range	−3 to 6 A	Negative for charging, positive for discharge
Cell voltage range	3.0 to 4.2 V	Cell terminal voltage constraint
Material ranges	x_n : 0.01 to 0.95; x_p : 0.20 to 0.95	Constraints on mean, center and surface fractions; not equivalent to the usable SOC interval 0 to 1.

Unless explicitly overridden, cases use this table and the material parameters in section 6.1. Illumination and idealized cycles are controlled inputs, not actual ISS conditions. Low and high temperature refer to specified battery temperatures.

In input tables, i_{min} and i_{max} are cell charge and discharge current bounds in A; v_{min} and v_{max} are cell voltage limits in V; η is distribution efficiency. Logical flags use 1 for true and 0 for false. Exact inputs are retained in suite.json; displayed table values may be rounded.

6 Test Data

6.1 Battery Material Parameters

Table 68. Battery Material Parameters

Parameter	Negative electrode	Positive electrode
Particle radius m	6.00e-06	5.00e-06
Active material volume fraction	0.75	0.665
Effective electrode area m ²	0.06	0.06
Electrode thickness m	8.50e-05	7.50e-05
Maximum lithium concentration mol/m ³	31000	63000
Reference diffusion coefficient m ² /s	3.30e-14	1.00e-14
Reference exchange current A	2.5	3
Diffusion activation energy J/mol	30300	25000
Reaction activation energy J/mol	35000	17500
Equilibrium potential polynomial coefficients V	0.65, -1.6, 2, -0.95	4.6, -1.3, 0.9, -0.8
Temperature polynomial coefficients V/K	1.20e-04, -3.00e-04, 1.50e-04, 0	-6.00e-05, 4.00e-05, 0, 0

Polynomial coefficients run from constant term to highest power, with reference temperature 298.15 K. Electrode charge is $Q_k = F \epsilon_k A_{el} L_k c_{k,max}$, the charge for a fraction change from 0 to 1 rather than a directly specified rated Ah capacity. These are existing illustrative test values, not a selected or calibrated cell.

Sources and constraints

Power/SDTwin_Power_Design_Report_CN.docx;

Power/SDTwin_Power_Design_Report_EN.docx; Thermal/sdtwin_sim/power_stand_in.py, example_power_records.

The material temperature range is 253.15 to 333.15 K. Cell ohmic loss is included in battery heat; distribution efficiency covers the PDU-to-Compute harness to avoid duplicate loss accounting.

6.2 Data Inventory and Reproduction

Use C:/Workspace/SpaceDC as the working directory. Run the following sequence to regenerate fixtures, independent results, verdicts and reports.

```
Thermal/.venv/Scripts/python.exe Power/verification/prepare.py
```

```
Thermal/.venv/Scripts/python.exe Power/verification/run_python.py
```

```
matlab -batch "addpath('Power/verification/matlab'); run_suite"
```

```
Thermal/.venv/Scripts/python.exe Power/verification/compare.py
```

```
python Power/verification/build_report.py
```

Use a document runtime with python-docx and matplotlib for the last command. Replace matlab with the full local R2026b executable path when needed.

Table 69. Data Inventory and Reproduction

File	Content
results/suite.json	Complete parameters, cases and criteria
matlab/sdtwin_power_reference.slx	Executed Simulink model
matlab/power_reference.m	Independent reference equations
results/<ID>_python.csv / <ID>_simulink.csv	Raw results from both implementations
results/<ID>_difference.csv	Direct differences at shared samples
results/<ID>.json / summary.json	All checks, errors and verdicts
results/<ID>_*_events.json	Actual event times and connectivity
results/NU-001_*_fine.csv	Independent refinement runs
results/history	Original inputs and criteria before coverage expansion

Input SHA256: f2d46a239842d0bba55d537efc8a793bb2c4df4644c4ca461afc9058a2400757

The implementation uses MathWorks Level 2 MATLAB S-function interfaces. The Simulink engine manages continuous states through Derivatives and InitializeConditions callbacks.

<https://www.mathworks.com/help/simulink/sfg/writing-level-2-matlab-s-functions.html>

6.3 Integrated Energy Comparison Data

Integrate each output series with the trapezoidal rule. Left and right event values have identical time, so their interval contributes zero energy. This compares energy at the common output cadence, not additional measured energy.

DY-002

Table 70. Integrated Energy Comparison Data

Energy source	Project J	Simulink J	Relative difference
Battery terminal power	-718286	-718286	7.62e-07
Delivered load power	1.08e+06	1.08e+06	1.36e-14
Actual solar power	1.85513e+06	1.85513e+06	2.95e-07
Pack heat	31993.8	31993.8	5.84e-07
Distribution loss	56842.1	56842.1	1.40e-14

DY-007

Table 71. Integrated Energy Comparison Data

Energy source	Project J	Simulink J	Relative difference
Battery terminal power	-768115	-768114	1.23e-06
Delivered load power	3.24e+06	3.24e+06	1.03e-14
Actual solar power	4.17864e+06	4.17864e+06	2.25e-07
Pack heat	58964.7	58964.7	4.62e-07
Distribution loss	170526	170526	1.08e-14

Battery terminal energy is signed: positive for net discharge and negative for net charge. Battery heat and distribution loss are integrated separately; internal heat is not added to terminal-power balance. Passing energy comparison does not change the failed SOC acceptance of these cases.

6.4 Numerical Convergence Data

NU-001 refines Python and Simulink separately. Each entry is the maximum absolute difference between baseline and refined runs of the same solver, not the difference between the two programs.

Table 72. Numerical Convergence Data

Solver	Quantity	Maximum change	Allowed change
python	State of charge fraction	1.55e-15	1.00e-07
python	Pack voltage V	1.78e-14	1.00e-05
python	Pack current A	4.44e-15	1.00e-05
python	Pack heat W	2.89e-14	1.00e-04
python	Battery temperature K	5.68e-13	1.00e-04
simulink	State of charge fraction	9.99e-16	1.00e-07
simulink	Pack voltage V	1.03e-13	1.00e-05
simulink	Pack current A	1.07e-14	1.00e-05
simulink	Pack heat W	3.02e-14	1.00e-04
simulink	Battery temperature K	1.02e-12	1.00e-04

Both solvers meet the refinement criterion of one tenth of the acceptance threshold. This check covers this smooth electrothermal scenario, not all stiff regimes or autonomous protection-event localization accuracy.

7 Test Summary

This report records the framework, inputs, criteria and individual comparisons of the Power model with independent Simulink simulation. All 20 numerical comparisons pass; 17 cases pass design acceptance and 3 fail due to the full-charge constraint.

Table 73. Power Module Test Summary

No	Case ID	Function	Verification Focus	Status	Result / Evidence
1	PV-001	solar_power	Solar irradiance and panel attitude	Pass	Completed 21 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/PV-001.json
2	BT-001	battery_response	Open circuit voltage and initial charge	Pass	Completed 9 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/BT-001.json
3	BT-002	battery_response	Charge and discharge voltage versus temperature	Pass	Completed 18 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/BT-002.json
4	BT-003	battery_response	Battery heat and reversible heat sign	Pass	Pack heat ranges from -0.0320434 to 3.23576 W; negative heat is retained. results/BT-003.json
5	BT-004	battery_response	Cell to series and parallel pack scaling	Pass	Completed 5 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/BT-004.json
6	PD-001	solve_power_allocation	Solar and battery power allocation	Pass	Completed 24 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/PD-001.json
7	PD-002	solve_power_allocation	Distribution efficiency and loss	Pass	Completed 9 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/PD-002.json
8	CT-001	solve_power_allocation	Charge limits and solar curtailment	Pass	Completed 6 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/CT-001.json
9	CT-002	solve_power_allocation	Discharge capability and supply shortfall	Pass	Completed 12 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/CT-002.json
10	CT-003	solve_power_allocation	Charging boundary at full state of charge	Fail	All three full-charge samples continue charging, with minimum cell current -0.544984 A. Numerical agreement passes, but the full-charge requirement fails. results/CT-003.json
11	PR-001	apply_power_event	Protection commands and latched states	Pass	Completed 8 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/PR-001.json
12	IV-001	PowerInputs / PowerState	Invalid inputs and state boundaries	Pass	Completed 6 input combinations. Output availability, discrete status and applicable numerical difference checks pass. results/IV-001.json
13	DY-001	solve_power_allocation / state integration	Charging under continuous sunlight	Pass	SOC changes from 65.517% to 88.654%; battery temperature changes from 293.150 K to 293.150 K. results/DY-001.json
14	DY-002	solve_power_allocation / state integration	Eclipse entry and exit	Fail	Peak SOC is 100.6153%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails. results/DY-002.json

No	Case ID	Function	Verification Focus	Status	Result / Evidence
15	DY-003	solve_power_allocation / state integration	Compute load steps	Pass	SOC changes from 65.517% to 71.390%; battery temperature changes from 293.150 K to 293.150 K. results/DY-003.json
16	DY-004	solve_power_allocation / state integration	Dynamic charge curtailment	Pass	SOC changes from 65.517% to 67.326%; battery temperature changes from 293.150 K to 293.150 K. results/DY-004.json
17	DY-005	solve_power_allocation / state integration	Supply loss latch and explicit restart	Pass	Supply trips at 100 s; the load remains disconnected after sunlight returns at 200 s. Explicit start at 300 s, stop at 400 s and restart at 500 s agree with expectations. results/DY-005.json
18	DY-006	solve_power_allocation / state integration	Battery electrothermal feedback	Pass	SOC changes from 65.517% to 44.282%; battery temperature changes from 283.150 K to 280.874 K. results/DY-006.json
19	DY-007	solve_power_allocation / state integration	Energy balance over three orbit cycles	Fail	Peak SOC is 104.2070%, above the configured 100% point. Both implementations exhibit this behavior, so overall acceptance fails. results/DY-007.json
20	NU-001	DOP853 / ode45	Solver convergence and accuracy review	Pass	SOC changes from 65.517% to 51.553%; battery temperature changes from 293.150 K to 291.345 K. results/NU-001.json

7.1 Anomalies and Conclusion

All 20 projects completed. Numerical agreement passes in 20 cases; design acceptance passes in 17 and fails in 3. The three failures expose the same omitted full-charge constraint in static and dynamic conditions.

Table 74. Anomalies and Verdicts

Case	Project result	Reason
CT-003	Charging at about minus 0.545 A per cell at SOC 1	No stop at the configured full-charge endpoint
DY-002	Peak SOC 100.6153%	Full charge exceeded in one cycle
DY-007	Peak SOC 104.2070%	Repeated cycles further expose the overcharge state

Code evidence: `_linear_limit` checks current and particle surface or center fractions, and `_search` additionally checks terminal voltage. The `xn,100` endpoint used to compute SOC is absent from charging limits. The independently implemented Simulink equations reproduce this behavior. Evidence supports a mismatch between the constraint set and the full-charge requirement, rather than an integration error.

Recommended correction: explicitly decide that usable SOC endpoints are charge and discharge constraints, reallocate at accepted boundary events, and curtail surplus solar at full charge. Rerun CT-003, DY-002, DY-007 and the regression suite after correction. Clipping displayed SOC to 1 is insufficient.

Overall Conclusion: ☐ Pass ☒ Fail ☐ Not Executed. All 20 numerical comparisons pass; three design-behavior cases fail acceptance.

On-site Test Expert Signature: _____ Date: _____